

Effective Independence: Quantifying the Reliability Ceiling of Same-Base-Model LLM Ensembles

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Abstract—Debate, self-consistency, and LLM-as-judge pipelines assume ensemble diversity they do not actually have when every voter is sampled from the same base model. We apply the classical effective-sample-size statistic N_{eff} to this setting and measure it across four diversity axes commonly used to build LLM ensembles (prompt, temperature, model size, and model family) on three reasoning and factuality benchmarks: GPQA-Diamond, MATH levels 4-5, and SimpleQA. Measured with the usual correlation coefficient, even $N = 3$ ensembles behave like barely more than a single independent voter ($N_{\text{eff}} = 1.09$ for prompt diversity, 1.51 for model-family diversity, 1.49 for all 12 configurations together). We then show that this coefficient flatters such ensembles, because it is bounded above by a function of the two members’ error rates: 11 of our 12 within-axis pairs sit exactly at that bound. Looking at the error sets directly explains why. In 53 of 66 configuration pairs, one member’s errors are a strict *subset* of the other’s, so the weaker member recovers nothing the stronger one missed and a majority vote has no material to work with. That nesting is total for pairs drawn from a single base model (28 of 28) and falls to about half for pairs spanning different labs (11 of 21), against a permutation baseline of roughly 15% ($p < 0.0005$). The consequence is measurable at the level practitioners care about: majority voting fails to beat the single best member of its own ensemble on every axis we tested, and trails it by 12.5 points across all 12 configurations at twelve times the API cost. We recommend reporting the fraction of ensemble member pairs whose errors are nested alongside any correlation-based diversity claim, since it is a direct count that cannot be inflated by adding a weaker member.

Index Terms—large language models, ensemble diversity, self-consistency, LLM-as-judge, multi-agent debate, effective sample size, reliability

I. INTRODUCTION

A. Background and Motivation

Self-consistency decoding, multi-agent debate, and LLM-as-judge ensembles all treat each sampled response as an independent vote. In practice, many of these pipelines draw every sample from the same underlying base model, varying only a shallow surface property such as the system prompt or the sampling temperature. If those samples share correlated failure modes, the effective diversity of the ensemble is far smaller than its nominal size N suggests. Majority voting, confidence aggregation, and judge panels then inherit a reliability ceiling that is invisible to the people building them.

B. Challenges in Quantifying Ensemble Diversity

Diversity is easy to induce along superficial axes (a different persona in the system prompt, a higher decoding temperature) and hard to induce along axes that plausibly decorrelate errors (a genuinely different training run, a different base model family). Prior evaluation practice rarely distinguishes between these axes’ contribution to ensemble reliability, and error correlation across ensemble members is rarely reported at all, leaving practitioners without a way to tell whether adding a 4th or 8th sample from the same model is buying real reliability or merely restating the 1st sample with different words.

C. Role of N_{eff} in This Work

We adapt the effective-sample-size framework used for correlated estimators to LLM ensembles. The quantity is Kish’s design effect [8], standard in survey sampling for discounting a nominal sample size by the correlation among its units: given N ensemble members with mean pairwise error correlation $\bar{\rho}$,

$$N_{\text{eff}} = \frac{N}{1 + (N - 1)\bar{\rho}}. \quad (1)$$

N_{eff} answers a concrete question: how many *independent* voters does this ensemble behave like? It also lets us compare diversity axes on a common scale rather than only on raw accuracy. We compute N_{eff} for 12 configurations spanning 4 axes (3 configurations per axis: prompt-diverse, temperature-diverse, size-diverse, and family-diverse), with bootstrap confidence intervals over questions, and report where each axis’s N_{eff} saturates as N grows.

D. Contributions

- We decompose measured ensemble independence by *construction axis*, holding the statistic fixed and varying one practitioner-controllable choice at a time, rather than auditing diversity in aggregate.
- We show that the statistic normally used for this is biased in a specific, correctable direction. The phi coefficient between two members’ error vectors is bounded above by a function of their error rates (Eq. 2), so members that differ in accuracy are credited with independence they do

not have. 11 of our 12 within-axis pairs sit exactly at that bound.

- We identify the structure responsible: LLM ensemble errors are largely *nested* rather than overlapping. In 53 of 66 pairs one member’s errors are a strict subset of the other’s, including all 28 pairs drawn from a single base model, against a permutation baseline of roughly 15% ($p < 0.0005$). A nested pair has zero complementary errors, so no aggregation rule over it can recover a question, which is a stronger statement than any correlation value alone supports.
- We show the nesting rate falls monotonically as members become genuinely more different (100%, 82%, 52% for pairs sharing a model, sharing a lab, and spanning labs), which gives the practitioner guidance a correlation ranking only gestures at.
- We connect the structure to the outcome it predicts. Majority voting never beats the best member of its own ensemble on any axis, trailing it by up to 25 points, and the number of questions a better aggregation rule could have recovered is between 0 and 2 per ensemble.
- We compare five allocation strategies at equal API cost under all three measures, and show that the ranking a correlation-based N_{eff} would suggest does not survive the correction (a 38% apparent advantage for cross-family ensembling becomes about 1%). We release the harness, the graded error matrix, and a decision log recording every deviation from the frozen specification.

II. RELATED WORK

Self-consistency decoding [1] samples multiple chain-of-thought [2] completions from a single model and takes a majority vote, implicitly treating the samples as independent draws. Multi-agent debate frameworks [3] extend this to multiple model instances arguing over several rounds, again typically instantiated with a single underlying model. LLM-as-judge evaluation [4] raises the same concern from the evaluation side: a judge panel built from one base model may share the judge’s own biases across every panelist. A parallel line of work builds explicit LLM ensembles for accuracy rather than for evaluation: Agent Forest [5] shows performance scaling purely from sampling-and-voting over more instances of the same model; Mixture-of-Agents [6] layers multiple LLMs so later agents condition on earlier ones’ outputs; and LLM-Blender [7] learns a pairwise ranker and fusion model over multiple LLMs’ candidate outputs. All of these methods’ reported gains are consistent with our central concern: none directly measures the pairwise error correlation their aggregation step relies on, so a reported improvement from adding more samples is not distinguishable, from the published results alone, from the diminishing-returns pattern we measure directly in Sec. IV. Our contribution is to make that correlation structure measurable via N_{eff} across the axes practitioners actually vary when they build such an ensemble.

Classical ensemble learning theory establishes that ensemble accuracy gains are governed by the trade-off between

individual member accuracy and pairwise error correlation: the ambiguity decomposition [9] first formalized this trade-off for regression ensembles, boosting [10] and bagging [11] are the two classical constructions for actively inducing decorrelated errors, and Dietterich [12] and Kuncheva and Whitaker [13] survey and compare diversity measures across this literature. N_{eff} (Eq. 1) is the same design-effect statistic used throughout this literature to summarize that trade-off in a single number. We apply it to a setting classical ensemble theory did not anticipate: API-only, black-box LLM ensembles, where individual members cannot be inspected, retrained, or explicitly decorrelated via a boosting- or bagging-style training procedure. They can only be reconfigured along the surface axes an API caller is given.

Two concurrent 2026 studies independently confirm that the independence assumption behind LLM ensembling is unreliable, from different angles. Kim [14] audits five diversity metrics across 31,900 ensemble subsets on MMLU-Pro and finds that apparent diversity gains are largely capability entanglement rather than genuine error decorrelation. Once capability is controlled for, simple majority voting beats the best individual model in under 10% of size-3 subsets. Ding [15] audits cross-model agreement as a confidence signal at large scale (265,000 samples on GPQA-Diamond and AIME) and reports only weak correlation (0.20–0.59) between agreement and correctness, concluding that self-consistency is a conditional proxy rather than a standalone confidence score. Both results corroborate our starting premise, that ensemble diversity is routinely overestimated, but neither decomposes *why*: they do not attribute the correlation they measure to specific, practitioner-controllable construction choices. Our contribution is complementary and narrower. Rather than auditing diversity metrics or agreement-as-confidence in general, we hold the classical design-effect statistic N_{eff} fixed and vary one structural axis at a time (prompt, temperature, size, family) to localize *where* an ensemble builder’s choices buy real independence. That is a decomposition, not a new headline claim that ensembles lack diversity; these two studies already establish that independently, and at far larger scale than our resource-constrained pilot.

We do, however, sharpen Kim’s [14] conclusion in a specific direction. That work attributes apparent diversity gains to capability entanglement, established by controlling for capability across ensemble subsets. Sec. IV-D identifies a mechanism that operates even within a single fixed subset: the correlation coefficient itself is bounded by the members’ error rates, so two members of unequal accuracy *cannot* record high correlation however similarly they fail. Capability and measured diversity are entangled not only through which models get selected, but arithmetically, inside the statistic. That makes the entanglement a property of the measurement rather than of the sampling, and it is correctable: Sec. IV-E recomputes every reported N_{eff} , per axis and per budget allocation, with an association measure that is not marginal-bounded.

III. METHODOLOGY

A. Datasets

We evaluate on three public benchmarks spanning multiple-choice and free-response formats: GPQA-Diamond [16], a graduate-level multiple-choice science benchmark; MATH [17] restricted to levels 4-5 (the hardest tier); and SimpleQA [18], a short-form factual QA benchmark. Free-tier API quotas (Sec. V) constrained the question set to $N = 30$ total (8 GPQA-Diamond, 11 MATH levels 4-5, 11 SimpleQA), a deliberate reduction from an original target of 800 questions. See `logs/decisions.md` for the two intermediate reductions (800→150→30) and the quota measurements that drove each.

B. Diversity Axes and Configurations

We define 12 configurations across 4 diversity axes, 3 configurations per axis, per the frozen specification in `config/project_spec.yaml`:

- **Prompt-diverse**: one fixed model at temperature 0.7, with three system-prompt personas (“careful analyst”, “confident expert”, “skeptical reviewer”).
- **Temperature-diverse**: the same fixed model and a neutral prompt, at temperatures $\{0.3, 0.7, 1.0\}$.
- **Size-diverse**: one model family, 3 sizes (small, medium, flagship), neutral prompt, temperature 0.7.
- **Family-diverse**: 3 different model providers/labs, neutral prompt, temperature 0.7.

Each (question, configuration) pair receives exactly one sample, since the goal is to measure cross-configuration error correlation rather than within-configuration sampling variance. One consequence is worth stating plainly: for the temperature axis, a single draw per configuration cannot separate temperature-induced diversity from ordinary sampling noise at that temperature, so that axis measures the diversity a practitioner actually gets from one sample per setting rather than the full sampling distribution.

C. Grading

Each dataset is graded by its own deterministic procedure. GPQA-Diamond uses exact match on the extracted answer letter. MATH uses symbolic equivalence via `math_verify` on the extracted `\boxed{\}` expression, so that algebraically equal forms are not counted as disagreements. SimpleQA uses normalized string containment against the reference answer, with an LLM judge as a fallback only when containment fails.

That fallback deserves an explicit caveat, because the judge is `gemini-3.5-flash`, which is also the base model behind 8 of the 12 configurations. A model therefore participates in grading responses that include its own. We mitigate this three ways. The judge is a fallback rather than the primary path, every invocation is written to an audit log released with the paper, and, as a robustness check, we recompute the paper’s main result with SimpleQA excluded entirely (Sec. IV-E). We would not use a self-judging setup by choice; it is a consequence of the free-tier constraint that no second model with sufficient daily quota was available.

D. Effective Independence Metric

For any subset of configurations we compute the pairwise error-correlation matrix and N_{eff} via Eq. 1, with 95% confidence intervals from 1,000 bootstrap resamples over questions. We additionally report the fraction of questions on which $\geq 80\%$ of an ensemble agrees and is wrong. This confident-correlated-error rate is the failure mode majority voting cannot detect or correct. Finally, we compare a fixed set of five allocation strategies at a matched budget of $k = 3$ ensemble members (Sec. IV-G), which re-uses the cached results already generated for the axis comparison and issues no additional API calls.

E. Experimental Harness

The harness, the frozen specification, the model configuration, the graded error matrix, and the full decision log are released with this paper at <https://github.com/reinamercy/Effective-Independence>. File paths given below refer to that repository. The decision log in particular records every deviation from the original specification, in the order the deviations happened and with the reasoning at the time.

All generations are produced through a cached, rate-limited API client (one persisted response per unique model/prompt/temperature/seed tuple), so no (question, configuration) pair is ever re-queried. Every API call is logged with the provider-returned model version string, token counts, and cost. All generation uses the free-tier model set frozen in `config/models.yaml`. The prompt- and temperature-diverse axes both use `gemini-3.5-flash` [19], which is also the medium tier of size-diverse; the small and flagship tiers of size-diverse are `gemini-3.5-flash-lite` and `gemini-3.7-flash` [19]. Family-diverse combines `gemini-3.5-flash` [19], `z-ai/glm-5.2` [20], and `nvidia/nemotron-3-ultra-550b-a55b` [21]. (The OpenAI slot originally used `openai/gpt-oss-20b` but was replaced with `z-ai/glm-5.2` mid-project after OpenRouter permanently de-listed the former from its free tier; see the decision log.) Dataset loading, configuration construction, cached generation, per-dataset grading and N_{eff} analysis are independently rerunnable modules connected only through on-disk artifacts, so any stage can be re-run in isolation.

IV. RESULTS

A. Quantitative Analysis

All results are computed on a fixed dataset of 248 graded (question, configuration) cells, 68.9% of the 30×12 design, at an overall accuracy of 73.0%. The absent cells were not dropped for any property of their content: they are the cells a per-model daily API quota did not permit us to collect (Sec. V), so the missingness is driven by which model a configuration uses rather than by question difficulty or by the answers themselves. Because that still leaves coverage uneven across configurations, we report below both what the uneven data supports and what survives on the subset where every configuration is graded. Table I reports N_{eff}

per axis with bootstrap 95% confidence intervals. Prompt-diverse and temperature-diverse, the two axes that vary only a surface property of one fixed base model, show the highest mean pairwise error correlation ($\bar{\rho} = 0.87$ and 0.77) and collapse to $N_{\text{eff}} \approx 1.1$ – 1.2 regardless of their nominal size $N = 3$. Size-diverse (one family, three parameter scales) is only marginally better ($\bar{\rho} = 0.73$, $N_{\text{eff}} = 1.22$), consistent with `config/models.yaml`’s documented caveat that its “flagship” tier is a newer Flash-generation model rather than a genuinely larger one, so this axis likely understates what true parameter-scale diversity would buy. Family-diverse scores best on this metric: mean error correlation drops to $\bar{\rho} = 0.50$ and N_{eff} rises to 1.51 , still far from the nominal $N = 3$. We deliberately stop short of calling this a settled axis ranking. Under partial coverage the family-diverse advantage rests on its two thinnest configuration pairs (Sec. IV-C), and those same two pairs turn out to be constrained by a marginal bound that accounts for their low correlation on its own (Sec. IV-D). Extending this to the full 12-configuration ensemble, N_{eff} rises only from about 1.0 at $N = 2$ to 1.49 at $N = 12$: adding nine further configurations buys roughly half of one additional effective voter.

TABLE I
 N_{EFF} PER AXIS, BOOTSTRAP 95% CI, $N = 3$ PER AXIS (PRELIMINARY, 248/360 CELLS). n_{MIN} IS THE SMALLEST NUMBER OF QUESTIONS GRADED FOR BOTH MEMBERS OF ANY PAIR WITHIN THAT AXIS, I.E. THE DEPTH THE AXIS’S WEAKEST CORRELATION ESTIMATE ACTUALLY RESTS ON.

Axis	$\bar{\rho}$	N_{eff}	95% CI	n_{min}
Prompt-diverse	0.87	1.09	[1.00, 1.29]	26
Temperature-diverse	0.77	1.18	[1.00, 3.00]	9
Size-diverse	0.73	1.22	[1.03, 1.64]	9
Family-diverse	0.50	1.51	[1.06, 1.74]	8
All 12 configs	0.64	1.49	[1.16, 1.90]	8

We additionally measure the confident-correlated-error rate: the fraction of questions where $\geq 80\%$ of a ≥ 3 -member ensemble is simultaneously wrong. Across all 12 configurations this occurs on 5 of 30 eligible questions (16.7%); per axis it is highest for prompt-diverse (5/26, 19.2%), followed by family-diverse (1/8, 12.5%), and lowest and roughly equal for temperature- and size-diverse (1/9, 11.1% each on this partial sample). A majority vote over any of these ensembles would confidently return the wrong answer on these questions. Diversity in surface presentation does not protect against it. Per-configuration accuracy also varies almost as much within an axis as across axes (e.g. size-diverse ranges from 70% at `size_diverse_small` to 89% at `size_diverse_medium`). That gap matters for interpreting N_{eff} , because an axis can spread its members widely on accuracy and still have highly correlated errors. What N_{eff} responds to is *which* questions each member gets wrong, not how many.

B. Qualitative Analysis

Both prompt-diverse confident-correlated-errors in the current data are single-fact recall failures where all three personas

converge on the same wrong value, not merely the same wrong direction. For “*In which year was Alexandra Park declared a local nature reserve?*” (ground truth: 2013), the careful analyst, confident expert, and skeptical reviewer personas each answer 2004, and two of the three additionally add the identical unprompted disambiguation “*referring to Alexandra Park in Hastings, East Sussex*”. The three prompts are not making three independent attempts at the question. They are retrieving the same wrong association from the underlying model’s parametric memory, and the persona instruction changes only surface phrasing (bolding, hedging words) rather than the retrieved fact. A second case, reporting a specific numeric result from a cited research paper, shows the same pattern: two of the three personas produce the byte-identical fabricated figure 96.53% against a true value of 92.71%, while the third confabulates a nearby but distinct number (95.53%). That is consistent with one wrong retrieval expressed with different rounding noise, not three independent readings of the source. In both cases a majority vote or judge panel built from these three configurations would confidently return the wrong answer, and would do so *more* confidently than any single sample, since correlated agreement is indistinguishable from genuine consensus to an aggregator that sees only votes. These examples illustrate the mechanism behind the nesting result in Sec. IV-E: the personas share a retrieval, so one cannot succeed where another fails.

C. How Much Data Each Correlation Rests On

Because coverage is still partial and the missing cells are not distributed uniformly across configurations (Sec. V), each pairwise correlation is computed only over questions where *both* members of the pair have a graded result, and that overlap varies sharply across pairs. Seven of the twelve within-axis pairs currently rest on 8–9 shared questions, and two of those saturate at exactly $\rho = 1.00$, which over nine questions is far more plausibly a small-sample artifact than a literal claim that two configurations never once disagree. Five pairs are well sampled ($n \geq 26$), and those sit in a much narrower band, $\rho = 0.68$ – 0.91 .

This bears directly on the axis ranking above. Family-diverse’s advantage is driven by its two *thinnest* pairs ($n = 8$, $\rho = 0.38$; $n = 9$, $\rho = 0.40$), whereas its only well-sampled pair, the two open-weight models at $n = 27$, has $\rho = 0.71$. That is squarely in the same range as the well-sampled same-base-model pairs ($\rho = 0.68$ for size-diverse at $n = 30$, $\rho = 0.90$ for prompt-diverse at $n = 30$). Read strictly, then, family-diverse’s apparent advantage rests disproportionately on the pairs we have the least data for, and the gap between it and the other axes may narrow as the remaining cells fill in. Of the correlation-based results reported here, this is the one we expect to move most at full coverage, which is part of why Sec. IV-E turns to a structural measure that does not depend on estimating a coefficient at all.

D. Correlation Is Capped by Accuracy Differences

A second constraint limits what the correlation numbers can support, and it cuts deeper than sample size. The phi coefficient between two binary error vectors is bounded by their marginals. If configuration A is wrong on a fraction p of questions and B on a fraction q , with $p \leq q$, then even when A 's errors are a strict subset of B 's, which is the most concordant arrangement available,

$$\phi_{\max} = \sqrt{\frac{p(1-q)}{q(1-p)}}, \quad (2)$$

which equals 1 only when $p = q$. The bound follows from maximising $\phi = (n_{11}n_{00} - n_{10}n_{01})/\sqrt{pq(1-p)(1-q)}$ over the 2×2 tables compatible with fixed marginals: the numerator is largest when the smaller error set is contained in the larger, which drives one off-diagonal cell to zero and leaves Eq. 2. Two configurations with different accuracy therefore *cannot* record a high correlation, however similarly they fail. A low ϕ is evidence of genuine error decorrelation only when it sits appreciably below this ceiling.

Checking each observed correlation against its own ceiling changes how Table I should be read. Eleven of the twelve within-axis pairs sit exactly at their marginal-bound ceiling: their errors are already as correlated as their differing accuracy permits. This includes both pairs responsible for family-diverse's advantage, where one member is far more accurate than the other ($p = 0.12$ against $q = 0.50$, and $p = 0.11$ against $q = 0.44$). At those error rates Eq. 2 caps ϕ at 0.378 and 0.395, and those are precisely the observed values. The low correlation on this axis is thus explained by its members having unequal accuracy, not by their failing on complementary questions.

Exactly one of the twelve within-axis pairs falls clearly below its ceiling: the two open-weight models, at $\phi = 0.71$ against a bound of 0.86. That pair is also the best-sampled on its axis ($n = 27$), and among the within-axis pairs it is the only evidence of error decorrelation not attributable to an accuracy mismatch. (Sec. IV-E widens the view to all 66 pairs, where cross-lab pairs supply further such cases.) We therefore do not claim that cross-family ensembling decorrelates errors. The defensible reading is narrower: measured correlation is uniformly close to the maximum the data permits, and what separates the axes in Table I is mostly how unequal their members' accuracies are.

This does not undermine the N_{eff} values themselves, which describe the ensemble's actual correlation structure whatever produces it. It does undermine any causal reading of the axis ranking, and it is a caution for practitioners: an ensemble can post a lower $\bar{\rho}$ simply by including a weaker member, which raises N_{eff} without making the ensemble's majority vote more trustworthy.

E. Errors Are Nested, Not Complementary

The bound in Eq. 2 is tight precisely when one member's errors are a subset of the other's, so the finding that most

pairs sit at their ceiling is really a claim about the *structure* of the error sets. That structure can be counted directly, without estimating a coefficient at all, and doing so turns out to be both more robust at this sample size and more informative about what an ensemble can do.

Call a pair *nested* when one member's error set contains the other's, i.e. one of the two off-diagonal cells of the 2×2 error table is empty. Pairs in which either member has no errors at all over the shared questions are excluded rather than counted as nested, since such a member is trivially contained in anything and would inflate the rate. Nesting is the case that matters operationally: if A 's mistakes are a subset of B 's, then B never answers a question correctly that A gets wrong, so no aggregation rule over $\{A, B\}$ can recover anything A missed; Sec. IV-F tests that consequence directly against majority voting. The pair is ordered by capability rather than diversified by it. A pair with complementary errors on both sides, by contrast, gives a vote something to work with.

The result is stark. In 53 of 66 configuration pairs the errors are fully nested (bootstrap 95% CI over questions [62%, 98%]). Table II breaks this down by how genuinely different the two members are, and the rate falls monotonically: **all 28 pairs drawn from the same base model are nested** (Wilson 95% CI [88%, 100%]), against 82% for pairs sharing a lab but not a model ([59%, 94%]), and 52% for pairs spanning different labs ([32%, 72%]). A Cochran–Armitage test across the three ordered groups confirms the trend rather than leaving it to the eye ($z = -4.12$, $p < 10^{-4}$). Fig. 1 plots the rates.

One scoping caveat belongs here rather than only in Limitations. Eight of our twelve configurations run on `gemini-3.5-flash`, so the same-model row, which is the strongest result in the paper, characterises one model's behaviour observed under eight different configurations rather than a property established across many base models. It says that varying prompt, temperature or decoding around *this* model never produced a complementary error. Whether that generalises to other base models is an open question our model set cannot answer.

Sparse tables could produce empty off-diagonal cells by luck, so this rate means nothing without a baseline. We therefore permute, holding each configuration's error count and its set of graded questions fixed and shuffling only *which* of those questions it fails. That destroys any real shared failure mode while preserving the marginals and the coverage pattern exactly. Over 2,000 permutations the expected nesting rate is 15% (95% range [8%, 26%]) across all pairs and 16% (95% range [4%, 32%]) among same-model pairs. The observed 80% and 100% are roughly five times chance, and no permutation in either null reached the observed value ($p < 0.0005$). The nesting is a property of the models, not of the sample size.

A second control addresses the grading circularity noted in Sec. III-C. Excluding SimpleQA entirely, so that no cell in the analysis depends on an LLM judge that shares a base model with the configurations being graded, leaves 19 questions and 21 defined pairs. The overall nesting rate there is 86% (18 of 21) and same-model pairs remain fully nested (3 of 3). The

effect is not produced by the self-judged dataset, though the surviving subset is small enough that we rest the headline number on the full set and offer this only as a check on direction.

The first row is the paper’s sharpest single result. Across every prompt-diverse and temperature-diverse pair we measured, and every other pair that happens to share a base model, there is not one question the second configuration gets right that the first gets wrong. Changing the persona or the sampling temperature reorders and rewords the output, and it changes how often the model is wrong, but on this data it never contributes an independent success. This is a stronger and more directly checkable claim than $\bar{\rho} = 0.87$, and it does not depend on estimating a correlation from nine questions.

It also explains the earlier numbers rather than merely accompanying them. Nested pairs are exactly the pairs pinned to the Eq. 2 ceiling, so the N_{eff} spread across axes in Table I is largely tracking how unequal the members’ accuracies are, while the underlying dependence structure is close to total throughout. Recomputing N_{eff} with Yule’s Q , an association measure that is not marginal-bounded and that reaches 1 whenever an off-diagonal cell is empty, makes this explicit: every axis falls to $N_{\text{eff}} \approx 1.0$ (Table II, Fig. 2), and the 12-configuration ensemble to 1.04.

We are explicit about the status of that substitution. Eq. 1 is derived for a common pairwise *correlation*, and Q is an association measure on a different scale, so $N_{\text{eff}}(Q)$ is not a drop-in estimator with the same sampling theory behind it. We use it as a diagnostic: it answers what the design effect would be if members were dependent to the degree Q records, and its role here is to show that the ordering and magnitudes implied by ϕ do not survive a measure free of the marginal cap. The nesting counts, which involve no formula at all, carry the claim; $N_{\text{eff}}(Q)$ expresses it on the scale readers already use.

We flag one caveat on Q specifically. Because it saturates at exactly 1 whenever an off-diagonal cell is empty, it is sensitive to single questions in sparse tables: several of our pairs have only six or seven errors in total, so one flipped answer would move Q off 1. We therefore treat the nesting counts, which are a direct property of the data, as the primary evidence, and Q as the formal restatement of what nesting implies for N_{eff} . Both point the same way, and both point further than ϕ does.

F. Does Majority Voting Actually Help?

Everything above characterises the ensemble through a statistic or a structure. Neither is what a practitioner buys. The question they face is whether aggregating k samples answers more questions correctly than using one model, so we measure that directly. It requires no additional generation: majority vote is computed from the same graded cells, with an even split counted as incorrect. To keep the panel constant across questions, each ensemble is scored only on questions where all of its members are graded.

Table III reports the result, and it is one-sided. **Majority voting does not beat the single best member of its own**

TABLE II
ERROR NESTING AND MARGINAL-ROBUST N_{EFF} (PRELIMINARY, 248/360 CELLS). A PAIR IS NESTED WHEN ONE MEMBER’S ERRORS ARE A SUBSET OF THE OTHER’S, SO THE PAIR HAS NO COMPLEMENTARY ERRORS FOR A VOTE TO EXPLOIT.

Pair type	Nested pairs	Rate
Same exact model	28 / 28	100%
Same lab, different model	14 / 17	82%
Different lab	11 / 21	52%
All pairs	53 / 66	80%

Axis	$\bar{Q}$	$N_{\text{eff}}(Q)$	$N_{\text{eff}}(\phi)$
Prompt-diverse	1.00	1.00	1.09
Temperature-diverse	1.00	1.00	1.18
Size-diverse	1.00	1.00	1.22
Family-diverse	0.99	1.01	1.51
All 12 configs	0.96	1.04	1.49

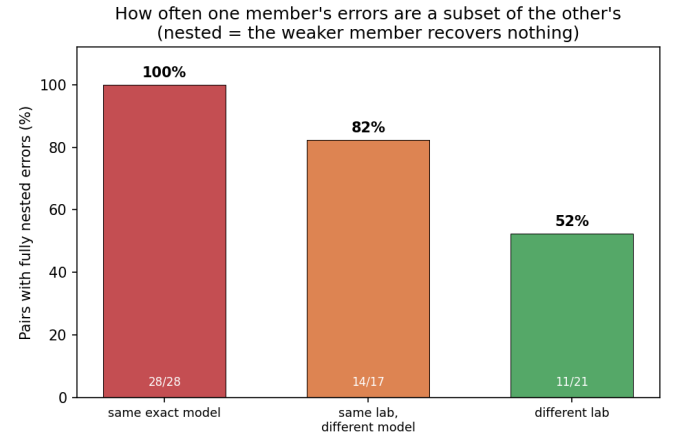


Fig. 1. How often one member’s errors are a strict subset of the other’s, by how genuinely different the two members are (preliminary, 248/360 cells).

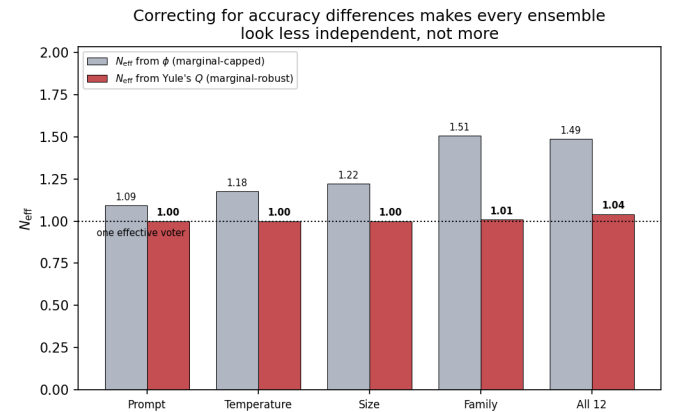


Fig. 2. N_{eff} computed with ϕ , which is capped by the members’ error rates, against N_{eff} computed with Yule’s Q , which is not (preliminary, 248/360 cells).

ensemble on any axis. It loses by 3.8 points on prompt-diverse, 11.1 on temperature-diverse and 25.0 on family-

diverse, and ties on size-diverse. Over all 12 configurations together it reaches 75.0% against 87.5% for the best member, a deficit of 12.5 points at twelve times the API cost.

This is the outcome the nesting structure predicts. When one member’s errors are a subset of another’s, the ensemble is ordered by capability, and a vote can only dilute the strongest member with weaker ones that contribute no question it missed. The recoverable column makes the size of the missed opportunity explicit: across all five ensembles, the number of questions where the majority was wrong while some member was right is 0 to 2. There is very little for a better aggregation rule to find, so the failure is not a deficiency of majority voting specifically. It is the absence of complementary errors to aggregate.

We note the obvious objection: the best member is only identifiable in hindsight, using the labels. A practitioner choosing a configuration in advance would more fairly be compared against the *mean* member, which voting does beat or match everywhere except family-diverse. The honest summary is therefore narrower than “voting is useless”: voting recovers roughly the average member’s accuracy, not the best member’s, and on this data it never converts k samples into an accuracy no individual member already had.

TABLE III
MAJORITY VOTE AGAINST THE MEMBERS IT AGGREGATES (248/360 CELLS). “RECOV.” COUNTS QUESTIONS WHERE THE MAJORITY WAS WRONG BUT AT LEAST ONE MEMBER WAS RIGHT. A NON-POSITIVE VOTE—BEST COLUMN MEANS VOTING NEVER BEAT THE STRONGEST MEMBER.

Ensemble	n_q	Vote	Best	Mean	Recov.
Prompt-diverse	26	76.9	80.8	76.9	1
Temperature-diverse	9	77.8	88.9	81.5	1
Size-diverse	9	88.9	88.9	81.5	0
Family-diverse	8	62.5	87.5	62.5	2
All 12 configs	8	75.0	87.5	71.9	1

G. Budget-Matched Allocation

The axis comparison above asks which kind of diversity decorrelates errors. The question a practitioner actually faces is narrower: given a fixed number of API calls per question, how should they be spent? We therefore compare five allocation strategies at a matched budget of $k = 3$ calls. Every strategy issues exactly the same number of requests, so any difference in N_{eff} is attributable purely to how the budget was allocated rather than to how large it was.

We note explicitly that the frozen specification framed this comparison as one at matched total API *cost*. Every model in the final configuration is free-tier, so dollar cost cannot be the matching variable; we match on k instead, which is both the genuinely scarce resource in this project (per-model daily request quotas) and the quantity a practitioner controls. This substitution is recorded in `logs/decisions.md` rather than made silently.

Table IV reports each allocation under all three measures. Reporting only the ϕ column here would contradict Sec. IV-D,

and the disagreement between the columns is large enough that it changes the conclusion.

Read through ϕ , cross-family looks decisively best: $N_{\text{eff}} = 1.51$ against 1.09 for three prompts of one model, a 38% gain at identical cost. Almost all of that gap is the accuracy artifact. Under Yule’s Q the same comparison is 1.01 against 1.00, a difference of about one percent, because cross-family’s apparent advantage came from its members having very unequal accuracy rather than from failing on different questions. We therefore do not claim a 38% reliability gain from crossing model families, and a practitioner should not budget for one.

What does survive the correction is categorical rather than quantitative, and it is visible in the last column. Every allocation built from a single base model is fully nested, 3 pairs of 3: no member of those ensembles ever answers a question that its partners miss. The two allocations that span more than one model each contain exactly one pair with complementary errors. That is the real distinction at a fixed budget. It is the difference between an ensemble that can in principle recover something and one that provably cannot, not a 38% improvement in a summary statistic.

This also corrects a reading the ϕ column invites. Under ϕ the mixed allocation (1.17) looks much worse than cross-family (1.51), which would suggest that spreading a budget across several axes is a poor substitute for changing model family. Under Q the two are indistinguishable (1.010 and 1.010), and both contain one complementary pair. The honest statement is that what matters is whether the allocation spans more than one base model at all, not how many different *kinds* of diversity it nominally mixes. With five allocations, three effective ties and $N = 30$ questions, we do not have the resolution to rank the non-nested allocations against each other, and we do not attempt to.

TABLE IV
ALLOCATIONS AT A MATCHED BUDGET OF $k = 3$ API CALLS PER QUESTION (PRELIMINARY, 248/360 CELLS). EVERY ROW COSTS THE SAME. THE ϕ COLUMN OVERSTATES THE SPREAD BECAUSE IT IS CAPPED BY DIFFERENCES IN MEMBER ACCURACY (SEC. IV-D); THE Q AND NESTING COLUMNS ARE THE ONES TO READ.

Allocation ($k = 3$)	$N_{\text{eff}}(\phi)$	$N_{\text{eff}}(Q)$	Nested
Same model, 3 prompts	1.09	1.00	3/3
Same model, 3 temperatures	1.18	1.00	3/3
Same family, 3 sizes	1.22	1.00	3/3
Mixed, one per axis	1.17	1.01	2/3
3 different families	1.51	1.01	2/3

V. LIMITATIONS

- **Model set:** budget constraints limited the model set to free-tier providers (Sec. III-E), rather than the closed-frontier-model set (Claude/GPT/Gemini-Pro) originally planned. This was a full, mid-project model-set rewrite driven by the discovery that the project owner has no funded Anthropic account and that this project’s real observed free-tier quota (20 requests/day/model) is far below what public documentation reports; the released

decision log gives the complete rationale, including that no model literally branded “Pro” is free on this account, and that the family-diverse axis mixes closed-weight (Gemini) and open-weight (GLM, Nemotron) models rather than three closed frontier labs’ flagships. Its members are also the most unequal in accuracy of any axis, which interacts directly with the bound below.

- **Dataset size:** at $N = 30$ total questions ($\sim 8\text{--}11$ per dataset), some axes’ 95% bootstrap confidence intervals on N_{eff} still span nearly their entire possible range (e.g. temperature-diverse: $[1.00, 3.00]$ against a nominal maximum of 3). This is a real statistical-power limitation rather than an artifact of the method, and results should be read as a pilot rather than a definitive ranking of axes. The intervals do tighten as coverage grows (prompt-diverse went from $[1.00, 3.00]$ at 205/360 cells to $[1.00, 1.29]$ at 248/360), but not every interval will become informative at $N = 30$.
- **Uneven coverage across configurations:** the missing cells are concentrated in the eight configurations sharing one model’s daily quota, so per-configuration graded counts range from 9 to 30. Correlations use pairwise-complete observations, so different axes are effectively measured on different question subsets and most pairs rest on only 8–9 shared questions (Sec. IV-C). Cross-axis comparisons are therefore confounded with coverage. A complete-case analysis removes that confound but leaves only 8 of 30 questions with all twelve configurations graded, which is too few to rank four axes against one another; we report the all-12 voting result on exactly those 8 questions in Table III and otherwise decline to rank. The axis ordering in Table I should be read as descriptive of this dataset rather than as an estimate of a population ordering. This is a property of the partial run rather than of the method: at full coverage every pair is measured on the same 30 questions.
- **Correlation is bounded by member accuracy:** ϕ cannot exceed the ceiling in Eq. 2 set by the two configurations’ error rates, and 11 of our 12 within-axis pairs currently sit exactly at that ceiling (Sec. IV-D). Differences in $\bar{\rho}$ across axes therefore largely track differences in how unequal the members’ accuracies are, and cannot be read as measuring how independently the members fail. A design that held member accuracy roughly constant across axes would separate the two effects; ours does not, because the free-tier model set fixes which models are available at each tier. We report the nesting counts (Sec. IV-E) partly because they are immune to this problem. Q , conversely, saturates at exactly 1 whenever an off-diagonal cell is empty, and several pairs have only six or seven errors in total, so one flipped answer would move it; it should be read as the formal consequence of the nesting we observe rather than as an independently precise estimate.
- **Nesting is measured per pair, not per ensemble:** a set of three configurations can be pairwise nested in a chain

without any single member dominating all others, and our counts do not distinguish those cases. The operational claim we make is the pairwise one.

- **Single sample per cell:** our design measures cross-configuration correlation, not within-configuration sampling variance, so N_{eff} estimates for a single axis at fixed N do not capture additional diversity available from resampling the same configuration. For the temperature axis in particular, one draw per setting cannot separate temperature-induced diversity from sampling noise.
- **A grading judge shares a base model with the configurations:** SimpleQA falls back to an LLM judge when string containment fails, and that judge is `gemini-3.5-flash`, which underlies 8 of the 12 configurations (Sec. III-C). Excluding SimpleQA leaves the nesting result intact (86% overall, same-model pairs still fully nested), so we do not believe it drives the finding, but a fully independent grader would be preferable and was not available within the free-tier quota.

VI. CONCLUSION

On the data collected here, prompt- and temperature-diversity are the two cheapest and most commonly used axes in practice, since neither requires changing model or provider. They are also the two that buy the least real independence: $N_{\text{eff}} \approx 1.1\text{--}1.2$ at nominal $N = 3$. Model-family diversity is the axis on which measured error correlation falls furthest, but we explicitly do not claim the reliability gain that its ϕ -based N_{eff} of 1.51 implies. That margin rests on the pairs with the least data behind them (Sec. IV-C) and on pairs whose low correlation is already fully accounted for by their members’ unequal accuracy (Sec. IV-D); corrected for that, the same comparison is 1.01 against 1.00.

The ranking is the least robust part of these results. The sturdiest finding is structural, and it is visible without estimating any coefficient: in 53 of 66 configuration pairs, one member’s errors are a strict subset of the other’s, and that includes every pair drawn from the same base model. Where the errors are nested there is nothing for an aggregation rule to recover, which is why a nominal three or twelve votes behaves like between one and one and a half, and like almost exactly one once the accuracy confound is removed. The prediction that follows is borne out directly: majority voting never beat the best member of its own ensemble, and only 0 to 2 questions per ensemble were even available for a smarter aggregator to rescue.

For debate, self-consistency, and judge-panel pipeline design, that means scaling the number of same-base-model samples is a much weaker reliability lever than its nominal N suggests. At a fixed budget the distinction our data supports is categorical rather than a matter of degree: allocations drawn entirely from one base model are fully nested and can recover nothing, while allocations spanning more than one base model contain at least one pair with complementary errors. How many different *kinds* of diversity are mixed matters less than whether the base model changes at all.

Measuring that structure carries its own trap: because ϕ is capped by the members’ error rates, an ensemble can report a flattering N_{eff} merely by including a weaker member. The remedy needs no extra API calls. Alongside any reported $\bar{\rho}$ or N_{eff} , publish the fraction of member pairs whose errors are nested. It is a direct count, it cannot be inflated by adding a weak member, and it answers what an ensemble builder actually needs to know: whether any member contributes a success the others miss.

These are pilot-scale results (Sec. V), consistent in direction with classical ensemble theory and with concurrent independent work [14], [15]. The magnitudes, and the axis ranking especially, remain provisional until coverage is complete. The nesting result is the one we expect to survive, since further coverage can only add questions on which a member might differ, and 28 of 28 same-model pairs have produced none so far.

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